

Use of Cython

Cython is a programming language based on Python. It has additional syntax for optional static-type declarations. The language allows the power of Python and C interchangeably in the same code. Apart from creating Python wrappers around natively-compiled code, Cython dramatically speeds up Python code. SageMath software is written in Python and Cython.

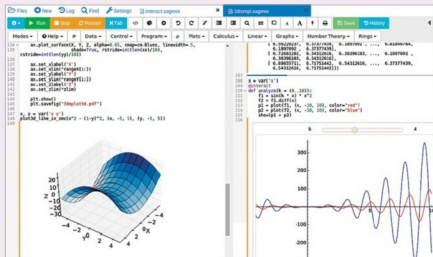


Fig. 2: Using SageMath, data can be represented in two and three dimensions

Additional functionalities are written as a new code and everything is glued together by a common interface.

Instead of inventing a new language for interfacing, the software makes use of Python as the gluing language to interact with all components. Python, being a well-known language in the scientific research community, is used to make the interface less complicated.

Necessity, the mother of all invention

The initial release of the software was made in early 2005. Whenever the scientific community felt the need for software to suit a particular mathematical application, the developers strictly followed the concept of release-early, release-often concept. This is the reason why there were more than 300 releases of the software in the last decade.

Reviews about the software in various forums suggest that developers' hard work has really paid off. "SageMath is easy to use and very

powerful. And I like the server-client principle behind it. Go on with the great work, SageMath motivated me to take the step from C to Python in just a few hours," comments Patrick Hammer, a user of the software.

"Sage is a work of art! It has a very clean syntax, is extremely well-documented and the coding is very clean. An average Python programmer like myself could create new functions in Sage. I have always wanted something like MATLAB but the price tag is a bit well... pricey for a 12-year-old. SAGE fits the bill perfectly," says Dhairav Pandya, who was then just 12 years old.

Is Sage giving up the game to its open source rivals?

The answer is 'Definitely not!' It is true that there is a negative increase in the user community for the software. Some users have migrated to other open source alternatives. Still, SageMath software is immensely used among mathematicians and the scientific research community.

In one of the recent blogs, Prof.

Smith identifies the reasons for the fall in the number of users. First, installation is not that easy. Second, limited resources in terms of books and supplementing resources make its usage difficult. Third, the software is missing key functionality needed in support of undergraduate teaching.

Developers are constantly trying to sort these issues out and provide adequate solutions. In order to overcome the issues related to installation, a full Web application called SageMathCloud was developed.

Open access books have been written for undergraduate courses based on this software. The software has been made more suitable for undergraduate STEM (short for science, tech, engineering and maths) courses.

Shift in focus

Learning from the experience of diminishing popularity, developers of this software are now shifting their focus to the following three categories of users: engineers employed in various industries, researchers in mathematics who are now not using this software and students in STEM courses.

Among these, the most promising group is the students who take undergraduate STEM courses. For a student who studies electronics engineering, purchase of a computation tool like MATLAB is not that feasible. Here comes the relevance of an open source alternative like SageMath.

Moreover, open source tools allow access to the source code, which is invaluable when it comes to the learning experience.

Last, but not the least, in classrooms, you need not worry about state-of-art performance that the industry demands. Summing it all up, the shift in focus to cater to the academia more will be the game changer in favour of SageMath in the near future. **EFY**

The author is assistant professor, department of ECE at SETCEM, Thrissur